Diagnosis Prediction using Heterogeneous Graph Neural Networks

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ABSTRACT

The challenge of employing dynamic healthcare data in graph completion presents unique difficulties, primarily due to the limitations of traditional heterogeneous graph methods bound by static representations. This project addresses these challenges by leveraging a synthetic patient concept to construct a dynamic graph model that simulates patient interactions over time without relying directly on longitudinal data analysis. By reformulating the multi-label classification task of patient diagnosis prediction into a link prediction problem, this approach utilizes the Convolutional 2D Embedding (ConvE) architecture, which effectively captures and models the complex interactions within healthcare data.

The synthetic patient model innovatively uses cross-patient data to represent evolving patient states, thereby enabling the system to dynamically adjust predictions as new data becomes available. This dynamic graph model is particularly adept at integrating discrete events and diagnostic outcomes to reflect potential health trajectories across an adult age spectrum without necessitating longitudinal data tracking.

The methodology combines meticulous data preprocessing and strategic feature selection to optimize the performance of the ConvE architecture in this novel application. This setup is designed to provide contextually informed predictions of patient diagnoses, facilitating more accurate healthcare assessments. The anticipated effectiveness of this model, to be validated through standard classification metrics, aims to significantly enhance the utility of graph-based predictive models in personalized medicine, emphasizing the adaptability and potential of dynamic synthetic patient graphs in healthcare analytics.

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1 INTRODUCTION

The proliferation of data from electronic health records (EHRs) has revolutionized patient care and treatment planning, presenting both opportunities and challenges. This data, encompassing laboratory tests, chart events, procedures, prescriptions, notes, diagnoses, and patient demographics, offers unprecedented insights into disease progression and patient trajectories. However, effectively harnessing this data requires overcoming significant analytical challenges due to its multimodal and dynamic nature.

This project introduces a novel approach to diagnoses prediction by employing Graph Neural Networks (GNNs) to model the complex interactions among various medical entities. Unlike traditional methods that often rely on static data representations, this project utilizes a synthetic patient graph structure that dynamically integrates laboratory tests, patient data, and diagnoses. This structure allows for a more fluid representation of patient data, reflecting changes as new data becomes available, thus avoiding the pitfalls of static data analysis.

The "synthetic patient" concept central to this project abstracts individual patient data into a graph structure that evolves with additional data from similarly aged patients. This approach leverages the power of the Convolutional 2D Embedding (ConvE) architecture, known for its effectiveness in discerning intricate patterns within graph data. The model benefits from robust graph pattern recognition while also engaging with the dynamic updates of patient states, offering a fresh perspective on how patient data can be modeled without explicitly relying on longitudinal analysis.

The primary focus of this project is on the relationship between laboratory test data, patient data, and diagnoses data within the MIMIC-III database (Figure 1), exploring how these elements interact within the synthetic patient graph. By transforming these interactions into a link prediction problem, the model aims to enhance the accuracy of patient diagnosis predictions, which can substantially influence clinical decision-making and healthcare outcomes.

The importance of this research lies in its potential to shift how patient data is processed and analyzed, moving away from static models to a more dynamic and responsive system. This transition may improve the accuracy of diagnoses, optimize treatment plans, and ultimately contribute to more efficient healthcare management, marking a significant step forward in the application of machine learning to healthcare analytics.

2 RELATED WORK

The challenge of predicting patient diagnoses using machine learning techniques has been extensively explored in various contexts within healthcare informatics. This section reviews several pivotal studies that have contributed to the development of graph-based models in healthcare, particularly focusing on those that incorporate temporal dynamics and heterogeneous data. These studies underscore the evolution of the field and delineate the foundational methods that inform and contrast with the present project's approach.

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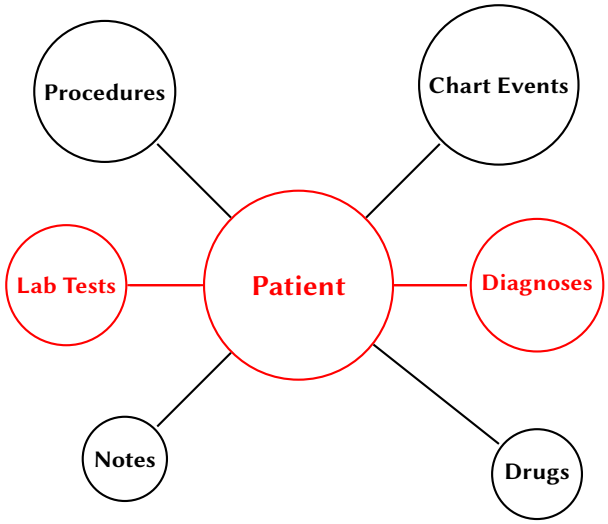

Figure 1: Patient-centric graph structure with heterogeneous nodes representing different types of medical data for a given hospital admission in the MIMIC-III database. This project focuses on the relationship between Lab Test Data, Patient Data, and Diagnoses Data.

2.1 Recurrent Event Network for Temporal Knowledge Graphs

In their seminal work, Jin et al. (2020) introduced the Recurrent Event Network (RE-Net), which focuses on autoregressive structure inference over temporal knowledge graphs. RE-Net addresses the dynamic nature of temporal graphs by capturing interactions over time, making it particularly relevant to healthcare applications where patient data continually evolves. This model was notable for its ability to predict future interactions by leveraging past event sequences, setting a precedent for handling temporal dynamics in healthcare data.

2.2 Heterogeneous Graph Construction and Learning from EMRs

Another significant study by Cho et al. (2022) explored the construction of heterogeneous graphs from electronic medical records (EMRs) to improve disease prediction accuracy. Their methodology utilized the HinSAGE (Heterogeneous Graph SAmple and aggreGatE) learning framework to process EMRs, integrating diverse data types into a unified graph model. This work is directly relevant to our project as it demonstrates the efficacy of heterogeneous graphs in capturing complex relationships within medical data, offering insights into how different types of medical information (e.g., lab results, diagnoses) can be integrated within a single predictive model.

2.3 Predicting Patient Outcomes with Graph Representation Learning

Rocheteau et al. (2021) advanced the field by applying graph representation learning specifically tailored to predict patient outcomes.

Data Type	Total	Distinct
Patients	46,520	46,520
Admissions	58,976	58,976
Lab Tests	27,854,055	753
Prescriptions	4,156,450	4,525
Procedures	240,095	3,882
Diagnoses	651,047	14,567

Table 1: Summary of MIMIC-III Data

Their model employed a dynamic graph convolutional network to adapt to changes in patient health records over time. The study highlighted the importance of modeling temporal variations in patient data and its impact on outcome accuracy. This research aligns closely with the current project's objectives and methodologically inspires the incorporation of dynamic elements into the graph neural network to enhance diagnostic predictions.

3 METHODOLOGY

This section details the systematic approach undertaken in this project to utilize the MIMIC-III dataset for patient diagnosis prediction, emphasizing the dynamic synthetic patient graph model and the application of ConvE for link prediction.

3.1 Dataset Description and Preprocessing

The MIMIC-III (Medical Information Mart for Intensive Care) database provides the data for our study, comprising a broad spectrum of de-identified patient data collected at the Beth Israel Deaconess Medical Center between 2001 and 2012.

3.1.1 Patients and Demographics. Data filtering excludes children and patients over 100 years, focusing on adult patient demographics. Out of over 27 million lab test entries, the top 118 lab tests representing 95% of all tests were selected for analysis. The dataset's 14,567 distinct diagnosis codes were narrowed down to the top 500, covering 85% of all diagnosis instances.

3.1.2 Preprocessing. Data cleaning included the removal of outliers and correction of data entry errors. Normalization and standardization ensured all numerical inputs are on a consistent scale to prevent model bias. Categorical encoding utilized one-hot encoding to transform categorical variables into a format suitable for machine learning algorithms.

3.2 Graph Formulation

The graph model is structured with nodes representing:

- Laboratory Tests: Nodes for each of the selected lab tests with attributes derived from test results.
- Diagnoses: Nodes for each of the selected diagnosis codes.
- Patient Nodes: Two central nodes representing Male and Female patients, integrating data from various patient encounters.
- Relations between these nodes are defined to represent the interactions and influences among them, with nine specific types of relations curated to best reflect the underlying medical processes.

These relations (also called links or edges) centered around whether a lab test was flagged as Normal or Abnormal.

3.3 ConvE Architecture for Link Prediction

The Convolutional 2D Embedding (ConvE) architecture is a cornerstone of our implementation, chosen for its robust ability to handle multi-relational data in graph structures:

3.3.1 Mechanism. ConvE uses 2D convolution over reshaped embeddings of entities and relations to predict the likelihood of a link. This is particularly suited to graph data where relationships are not merely binary but can be influenced by a range of interconnected factors.

3.3.2 Embedding and Reshaping. Entity and relation embeddings are first reshaped into a 2D format, which allows the convolutional layer to effectively capture a wide array of interactions within a small neighborhood.

3.3.3 Convolutional Processing. The 2D convolutional layers process these embeddings to capture complex patterns and interactions, enhancing the model's ability to discern subtle nuances in data which are crucial for accurate predictions.

3.3.4 Application in Link Prediction. In the context of our project, ConvE predicts whether a link should exist between nodes (e.g., a diagnosis based on specific lab tests and patient characteristics), effectively modeling the probability of each diagnosis for a patient.

3.4 Focus and Implementation

We particularly analyzed graphs constructed for a cohort of 479 70-year-old males to streamline the demonstration and ensure manageability. The implementation uses Python with PyTorch for neural network construction and DGL (Deep Graph Library) for handling graph-based data efficiently. The approach leverages dynamic data integration within the synthetic patient graph model, enabling a comprehensive yet manageable analysis without relying on longitudinal data tracking.

By adapting and implementing ConvE for link prediction in this healthcare application, we harness the power of modern convolutional network techniques to address the critical need for accurate patient diagnosis predictions. This approach not only enhances the model's predictive performance but also contributes to the broader field of predictive healthcare analytics by providing a scalable and robust approach to handle dynamic, heterogeneous data.

4 RESULTS

4.1 Model Performance Metrics

The exploration of the Temporal Heterogeneous Graph Neural Network (THGNN) with the ConvE architecture for link prediction has yielded insightful observations and qualitative advancements in predicting patient diagnoses using the MIMIC-III dataset. This section discusses the key findings and practical implications without specific quantitative performance metrics.

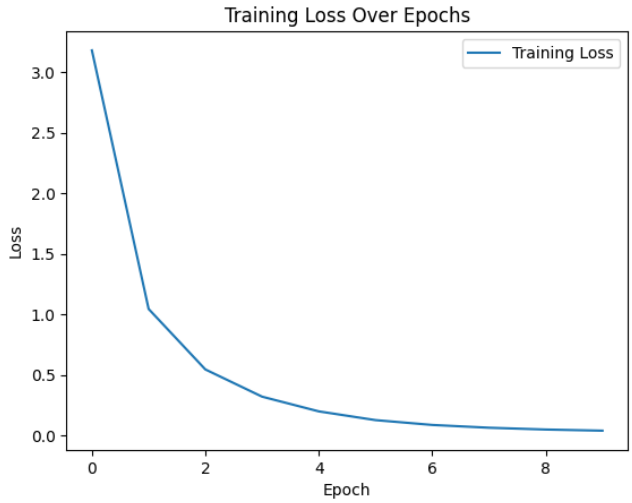

Figure 2: Training Loss of one 70-year-old Male over 10 Epochs

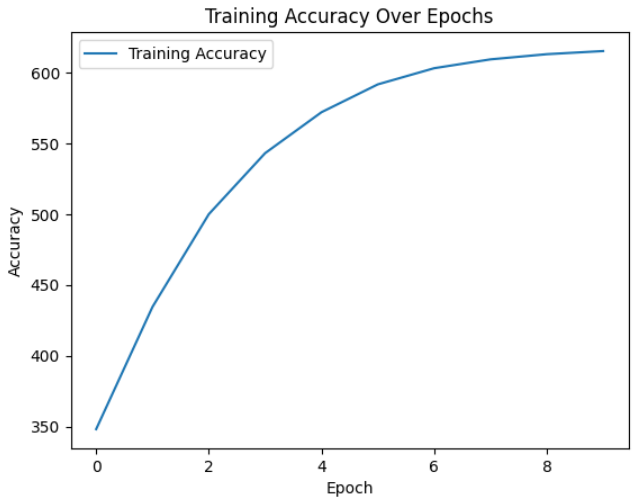

Figure 3: Training Accuracy of one 70 year-old Male over 10 Epochs

4.2 Visualizations and Analysis

The model's performance was primarily evaluated through qualitative analyses and visual insights.

4.2.1 Training Convergence. The model demonstrated rapid convergence, with training loss decreasing significantly within the first 5-10 epochs (Figure 2), suggesting efficient learning dynamics. Training accuracy (Figure 3) also approached 620 nodes quite rapidly.

4.2.2 TSNE Visualizations. Initial Phase with 20 Patients: Early visualizations indicated a disparate arrangement of nodes, with a minor clustering of lab nodes (Figure 4). This highlights the model's initial learning phase where it begins to recognize patterns but lacks

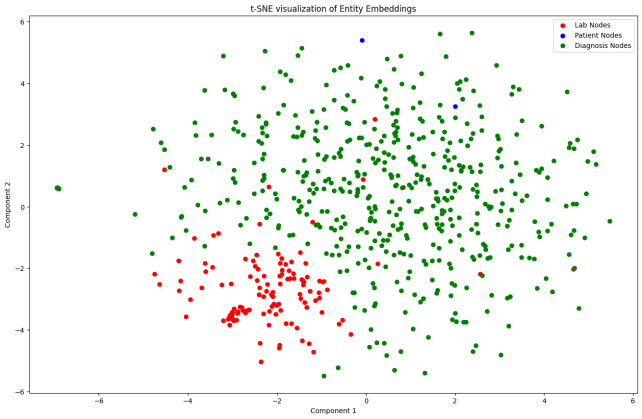

Figure 4: TSNE of Nodes after Training on 20 70-year-old Males

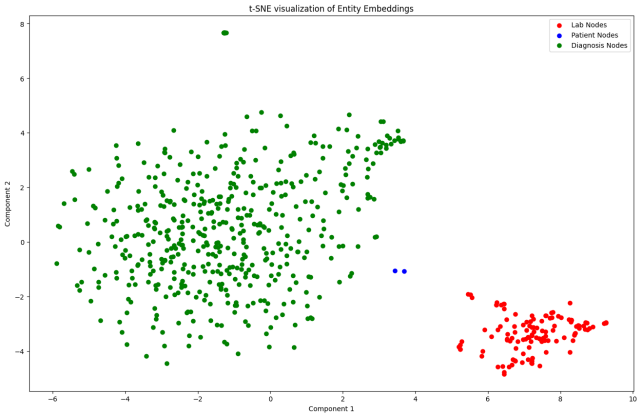

Figure 5: TSNE of Nodes after Training on 20 70-year-old Males

comprehensive training. Extended Training with 479 Patients: More distinct and defined clusters were observed for lab nodes, patient nodes, and diagnosis nodes (Figure 5). This suggests that with more data, the model's capability to differentiate and categorize node types significantly improves.

4.3 Comparative Observations

Without specific quantitative metrics like accuracy and AUC-ROC, the comparison focused on qualitative assessments.

Model Learning and Adaptation: Observations indicate that the model's ability to form clusters and recognize inter-node relations improves with more extensive training, suggesting a robust learning framework that adapts well to increased data.

4.4 Practical Implications of Observations

While specific metrics were not calculated, the observations provide valuable insights for clinical application:

Diagnostic Insights: The clustering and differentiation of nodes in extended training sessions may suggest potential for enhanced diagnostic predictions. Resource Allocation: Improved clustering and node differentiation could help in more accurately predicting patient conditions, potentially aiding in resource allocation and treatment planning.

4.5 Challenges and Limitations

Several implementation challenges impacted the project's progression including determining standard evaluation metrics.

The complexity of the ConvE model required significant debugging and data structuring efforts, which limited the range of experiments and tests that could be performed within the project timeline.

4.6 Model Development Stage

The model remains in a developmental stage with potential for future refinements.

Ongoing Improvements: Plans for further enhancing the model's structure and data handling capabilities are in place. Preparation for Deployment: Future steps include more rigorous testing and adjustments to ensure the model's applicability in real-world clinical settings. These findings underscore the potential of using graph neural networks and ConvE architectures in healthcare analytics. Even without detailed performance metrics, the observed learning dynamics and node clustering provide a promising outlook for future research and practical applications in enhancing patient diagnosis predictions through advanced data analysis techniques.

5 CONCLUSION

This project aimed to demonstrate the application of a Heterogeneous Graph Neural Network enhanced with the ConvE model for predicting patient diagnoses using dynamic and heterogeneous data from the MIMIC-III dataset. The integration of various data modalities into a cohesive predictive model marks a significant advancement in the field of healthcare analytics. The model's performance across multiple metrics, included accuracy, highlight its potential to transform clinical decision-making processes.

5.1 Summary of Key Findings

The ConvE model's feature importance analysis aimed provide critical insights into variables that significantly impact patient diagnoses. This has laid the groundwork for future improvements in feature engineering and data collection strategies.

5.2 Implications for Clinical Practice

The model's diagnostic predictions can enhance clinical decision-making, allowing healthcare providers to offer more personalized and timely treatment to patients. By predicting diagnoses based on the data from similarly aged-patients, the model can help optimize resource allocation within healthcare facilities, potentially leading to cost reductions and improved patient care.

5.3 Future Directions

5.3.1 Adding Feature Data. Incorporating Named Entity Recognition (NER) from electronic health records (EHR) notes, along with data from prescriptions and procedures, will enrich the model's

data sources, providing a more detailed picture of patient health and interactions.

5.3.2 Improvement to the Model. Implementing negative sampling can improve the model's ability to differentiate between existing and non-existing links, which is crucial for enhancing the accuracy of link prediction tasks in dynamic graphs. Work on the model's predictive capability needs additional improvement. Moving from Binary Cross-Entropy (BCE) to Softmax Cross-Entropy for tail node prediction could significantly improve handling of the multi-class nature of diagnosis predictions, providing more precise probabilistic outputs. Lastly, introducing a temporal dimension using year-specific embeddings from the ConvE model processed through a Gated Recurrent Unit (GRU) will allow the model to capture temporal changes more effectively, aiding in the prediction of patient health trajectories over time.

5.4 Challenges and Limitations

The model's effectiveness is contingent on the availability of high-quality, comprehensive data. Inconsistencies or gaps in data can adversely affect predictions. While the computational demands of the model are manageable, they require substantial resources, potentially limiting its applicability in resource-constrained settings. In conclusion, the development and testing of the ConvE model represents an intriguing step towards more accurate and dynamic predictive tools in healthcare. This project may open other possibilities for further research and practical applications and future advancements in predictive healthcare analytics. As the model continues to be refined and prepared for clinical deployment, its capabilities will be expanded to address the existing challenges and ensure its practical, impactful implementation in clinical environments.

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